

Object counting (Human counting)

Assignment Documentation

BMCS2133 Image Processing

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Programme : Bachelor of Data Science

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Introduction

1.1 Problem Background

In large-scale public events and high-density urban spaces, maintaining crowd safety is not only a logistical challenge but a critical factor in preventing accidents, injuries, and fatalities. Events such as concerts, sports matches, religious gatherings, and political rallies often draw thousands of people into confined venues, where poor crowd control can lead to catastrophic outcomes. These environments require continuous situational awareness and immediate responses to shifting crowd dynamics. Traditional manual crowd control methods—such as visual monitoring and static barriers—are often reactive, imprecise, and vulnerable to human error. This highlights the urgent need for a proactive and intelligent system that ensures crowd safety in real-time (Asif, 2023).

The project titled “Human Counting for Crowd Safety Management” addresses this gap by utilizing human-counting technologies to monitor and manage the number of individuals in high-density areas. By tracking real-time crowd movement and density, this system enables early interventions, facilitates emergency planning, and reduces the likelihood of overcrowding-related incidents. Crowd management is particularly vital when emotions are running high, such as during competitive sports events or politically charged demonstrations. In such cases, poorly managed crowds not only pose physical risks but also create psychological distress for attendees, staff, and first responders.

Poor crowd management can lead to tragic consequences, as evidenced in incidents such as the 2022 Itaewon crowd crush in South Korea. Survivors reported long-term trauma, and in some cases, security personnel were also severely affected. These cases underscore that ensuring crowd safety isn't just about physical well-being—psychological safety matters too. Moreover, effective crowd control contributes to the overall success and economic viability of events, encouraging repeat attendance and strengthening public trust in event organizers (Haghani et al., 2023).

This project aligns with multiple United Nations Sustainable Development Goals (SDGs). Specifically:

1. SDG 3: Good Health and Well-being – by reducing the risk of injuries and trauma in mass gatherings;
2. SDG 9: Industry, Innovation, and Infrastructure – through the integration of smart crowd-monitoring systems into urban and event infrastructure;
3. SDG 11: Sustainable Cities and Communities – by promoting safe, inclusive public spaces;
4. SDG 16: Peace, Justice, and Strong Institutions – by supporting safe environments where people can assemble freely without fear of harm.

By shifting from manual to automated, data-driven crowd monitoring, this system ensures safer environments and facilitates efficient emergency responses. It also reduces the burden on human personnel, allowing them to focus on high-priority interventions instead of routine surveillance. As urbanization and event attendance continue to rise globally, integrating human-counting technologies into public safety protocols is no longer optional—it's a necessity.

Keywords Explanation

Human Counting: Human counting refers to the use of technologies—such as video analytics, infrared sensors, or Wi-Fi tracking—to monitor and count individuals in a specific area. In crowd management, this ensures real-time awareness of how many people are in a venue or zone, helping to prevent overcrowding and support emergency decision-making.

Crowd Safety: Crowd safety encompasses strategies and measures designed to protect people in large gatherings from physical harm or distress. Effective crowd safety systems reduce the risk of stampedes, bottlenecks, and panic responses, ensuring the well-being of attendees, staff, and responders.

Event Management: Event management involves the planning, coordination, and execution of public or private events. A critical component of this is the ability to manage crowds

effectively, which requires real-time data, contingency planning, and technological support to ensure smooth operations and safety compliance.

1.2 Objectives/Aims

1. To achieve high accuracy in real-time human counting by enhancing the detection model's precision through algorithm improvements, high-quality data usage, and environmental calibration, thereby minimizing false positives and false negatives during live tracking.
2. To implement automated alert notifications that inform authorized personnel when the number of people exceeds predefined occupancy limits. Alerts triggered by threshold violations help reduce safety risks such as fire hazards and non-compliance with regulations.
3. To test the scalability of the human counting system across different layouts. The system is evaluated in various spatial configurations to assess its robustness and adaptability, ensuring consistent performance despite structural or environmental differences, thereby supporting broad applicability across diverse settings.

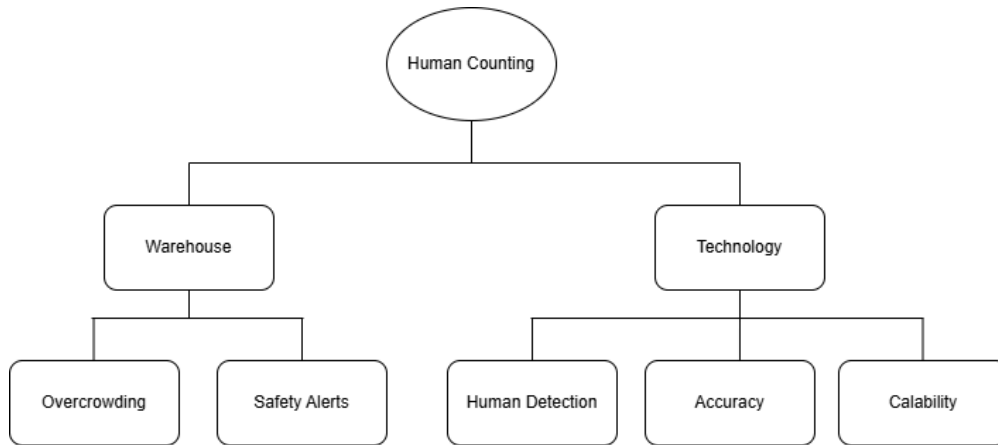
1.3 Motivation

- Commercial Perspective:
 1. Addresses a critical operational need for real-time crowd tracking at large public events.
 2. Highly marketable to event organizers, security agencies, local authorities, and smart city planners.
 3. Helps ensure compliance with public safety regulations (e.g., venue capacity limits, crowd control laws).
 4. Reduces the need for manual crowd supervision through automated alerts and real-time occupancy data.
 5. Enhances decision-making during events by providing live crowd metrics and early warning signals

- Social Perspective:
 1. Ensures safer environments by preventing overcrowding, stampedes, and delayed emergency responses.
 2. Supports social sustainability by prioritizing the safety and psychological well-being of attendees, staff, and performers.
 3. Offers contactless crowd management options adaptable for use during public health emergencies (e.g., COVID-19).
 4. Promotes public trust and confidence in attending mass gatherings by demonstrating responsible safety practices.
 5. Aligns with ethical and inclusive urban design principles as part of modern smart event infrastructure.
 6. Commercially valuable and socially impactful in building safer and more resilient public spaces.

Literature Review / Contemporary Study

2.1 Background of the applications



Deep learning is one of the popular topics in the field of machine learning. Deep learning algorithms have been successfully applied in different modalities and fields. Image processing, machine translation , and natural language processing are some of them. Deep learning is also used in other fields apart from these. For example, in recent years, deep learning algorithms are used for sound classification , language recognition, or cancer diagnosis as well.

There are two types of methods for person detection: Region of Interest (ROI) and Line of Interest (LOI). In the ROI-based method, the number of people in a limited region is estimated. In the LOI-based method, on the other hand, the number of people crossing a predetermined line is estimated. Deep learning-based models are used in the ROI-based system approach. Deep learning is frequently used in object detection and classification tasks. This situation encouraged researchers to use deep learning on people counting problems, as well. In this study, deep learning methods were used with an ROI-based approach (Gündüz & Işık, 2023).

Firstly, YOLO is known for its remarkable speed. By treating object detection as a regression problem, it eliminates the need for a complicated processing pipeline. At test time, the system

simply passes the input image through the neural network to generate predictions. The standard model can process 45 frames per second without batch processing on a Titan X GPU, while a faster version can exceed 150 frames per second. This enables real-time video processing with latency under 25 milliseconds. In addition to its speed, YOLO also achieves more than twice the mean average precision compared to other real-time detection systems.

Secondly, YOLO takes a holistic approach to image analysis. Unlike sliding window or region proposal-based methods, it processes the entire image during both training and testing, allowing it to capture contextual relationships between objects and their surroundings. As a result, YOLO avoids many of the mistakes made by other detectors, such as Fast R-CNN, which can misclassify background areas due to limited context. In fact, YOLO produces less than half as many background errors as Fast R-CNN.

Thirdly, YOLO demonstrates strong generalization capabilities. When trained on photographs and tested on artwork, it significantly outperforms leading detectors like DPM and R-CNN. This generalizability makes YOLO more robust when applied to new or unfamiliar data domains.

However, YOLO still falls short of the highest-accuracy detection models. While it excels at quickly identifying objects, it can have difficulty precisely locating them, particularly smaller ones. These trade-offs are further explored in the experimental section of the research (Redmon et al., 2016) .

2.1.1 Head Detection in Colleges and Universities

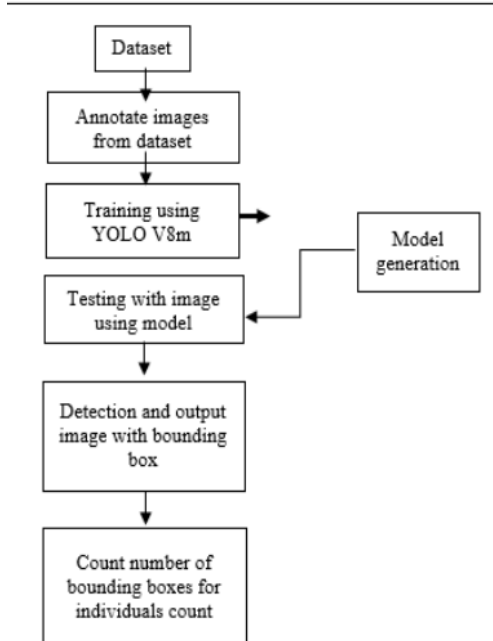
Many colleges and universities face the issue of insufficient dedicated self-study classrooms due to limited space and high classroom usage for scheduled courses, meetings, or events. As a result, students often use available vacant classrooms for self-study. However, they frequently encounter difficulties in determining which classrooms are currently available and how crowded they are. While many institutions in China have implemented classroom inquiry systems that allow students to check room availability based on scheduling and booking data via PC or mobile devices, these systems typically do not provide real-time information about how many students are currently occupying the rooms. This gap has led to growing demand—such as at China University of Geosciences (Beijing)—for solutions that allow students to check not only whether a room is free but also how crowded it is. Addressing this need has become increasingly urgent to better manage study space resources and improve the self-study experience.

The core of image-based population statistics lies in effectively representing human features and establishing clear standards for recognizing them. Depending on the specific targets of detection, several methods are commonly used: head detection, head and shoulder detection, face detection, combined head and face detection, body contour detection, and skin color detection. In this study, the focus is on counting students engaged in self-study within a classroom setting. In such environments, students typically have their heads lowered, making facial detection unreliable. However, their heads remain relatively visible and unobstructed, making head detection a more practical approach. Since the goal is not to achieve extremely high accuracy but to provide students with general guidance on available seating in study rooms, the head detection method is chosen and implemented accordingly (Liu & Yu, 2022).

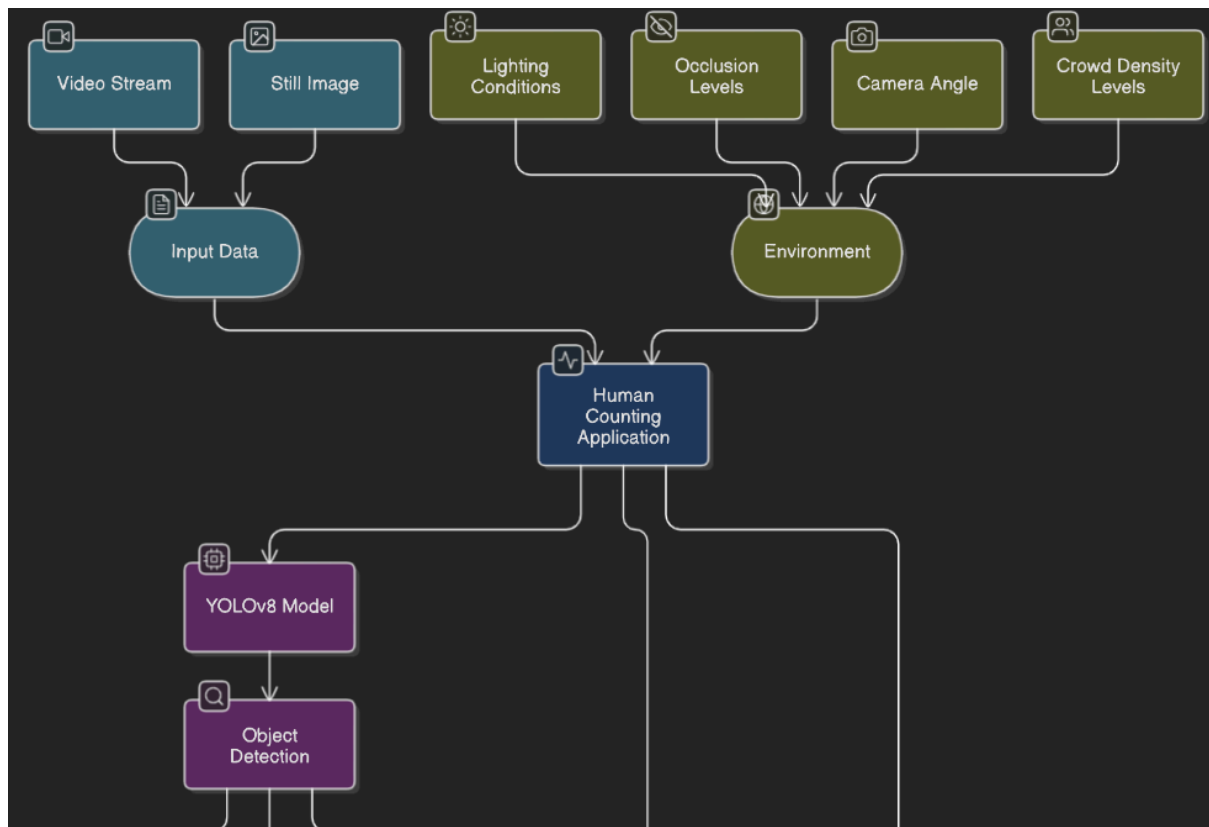
2.1.2 Counting Individuals in an Image using Machine Learning Technique

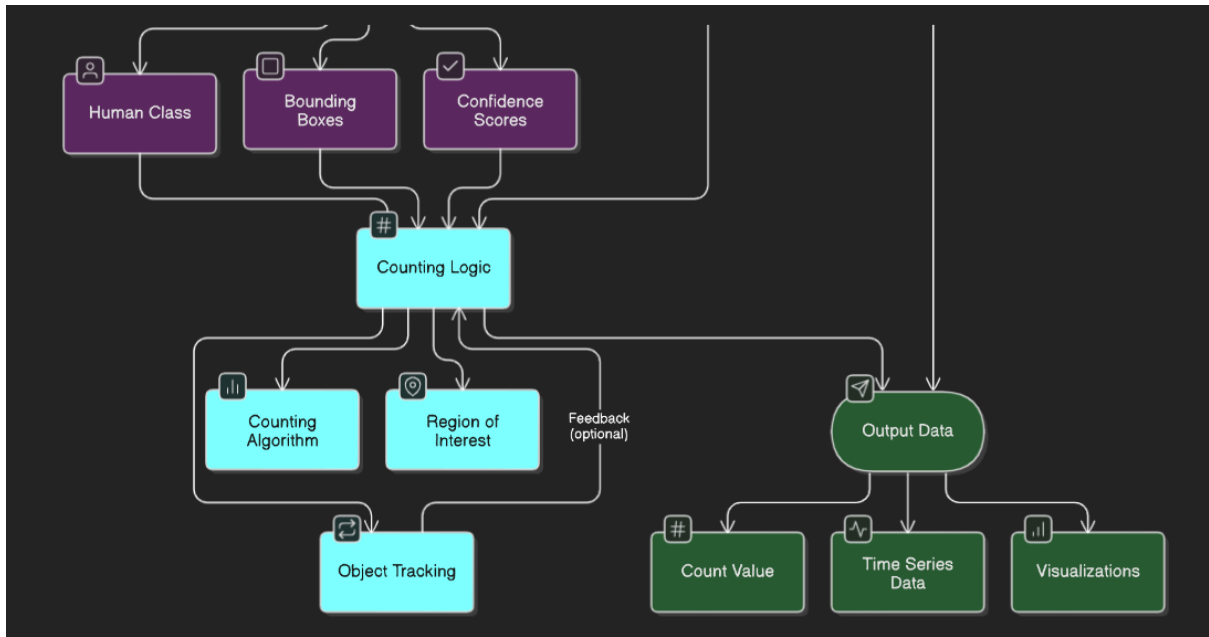
YOLO (You Only Look Once) is a well-known object detection framework recognised for its real-time performance and high accuracy. The architecture of YOLO consists of five primary stages: the input stage, a feature extraction Backbone, a Neck component, dense prediction, and a final sparse prediction layer. The Backbone employs a series of convolutional layers arranged in a pyramid structure to extract hierarchical features from the input image. These features are then processed by the Neck, which incorporates a Feature Pyramid Network (FPN) and Path Aggregation Network (PANet) to enhance feature fusion across scales. The dense prediction stage generates detection outputs for various image locations, while the sparse prediction stage refines these results, drawing precise bounding boxes around detected objects.

YOLOv8, particularly the YOLOv8-m (medium) variant, is a robust, open-source implementation widely used for tasks such as individual detection and counting. The detection pipeline begins with dataset preparation, which involves collecting and annotating images of individuals. Once trained, the model produces a pretrained weight file used for inference on new data. To improve output clarity, Non-Maximum Suppression (NMS) is applied to remove redundant overlapping bounding boxes. Each bounding box is assigned a class label, allowing accurate identification and counting of individuals. This framework has demonstrated effective performance in crowded environments and is suitable for real-time deployment in population analysis tasks (D P & Kumar, 2024).



2.1.3 Ontology Diagram





2.2 Analysis of selected tool/algorithm with any other relevant tools/algorithm

Tools comparison	Remark	Tool/Algorithm 1: YOLOv5 (You Only Look Once)	Tool/Algorithm 2: YOLOv8	Tool/Algorithm 3: YOLOv11
Type of licence and open source licence	State all types of licence	AGPL-3.0	AGPL-3.0	AGPL-3.0
Year founded	When is this tool being introduced?	2020	2023	2024
Founding company	Owner	Ultralytics	Ultralytics	Ultralytics
Licence Pricing	Compare the prices if the licence is used for development and business/commercialization	Free for development; commercial use may require paid license	Free for development; commercial use may require paid license	Free for development; commercial use may require paid license
Supported features	What features does it offer?	- Real-time object detection - High speed - Small model sizes	- Object detection - Instance segmentation - Pose estimation - Oriented bounding	- Transformer-based backbone - Unified model for detection, segmentation,

		(YOLOv5n) - Multi-platform support (ONNX, CoreML, TensorRT)	boxes (OBB) - Better model export support	and pose - Dual-label assignment - NMS-free training
Common applications	In what areas is this tool usually used?	- CCTV and surveillance - Retail analytics - Autonomous driving - Traffic monitoring	- Drone/aerial imagery analysis - Text detection in images - Factory automation - Security systems	- High-precision medical imaging - Advanced robotics - Smart cities - Complex real-time vision systems
Limitations	The drawbacks of the software	- Less effective with small or overlapping objects - Performance varies across hardware - Requires labeled datasets (Jiang et al., 2022)	- Higher GPU requirements - Training time increases due to added complexity (Nidya Chitrasingrum et al., 2024)	- Computationally expensive - Not beginner-friendly - Requires latest hardware for full potential (Kim, 2024)

2.3 Justify why the selected tool is suitable

The primary tool used in this project is YOLOv11, a state-of-the-art object detection algorithm developed by Ultralytics. It was selected to build a high-performance human detection system tailored for real-time monitoring and analysis in environments such as warehouses, public spaces, and security-sensitive areas.

Why YOLOv11?

1. Real-Time and Accurate Human Detection:

YOLOv11 introduces a transformer-based backbone, which significantly improves accuracy, especially for detecting humans in crowded, partially occluded, or low-light conditions. This is essential in human detection scenarios, where the system must reliably recognise individuals in constantly changing environments, such as people walking behind equipment, partially out of frame, or clustered in groups.

2. Multi-Task Support for Enhanced Analysis:

In addition to basic detection, YOLOv11 supports segmentation and pose estimation, allowing the system to not only detect people but also recognise body posture or movement. This is especially useful in safety-critical applications—such as identifying when a person has fallen, entered a restricted zone, or is not wearing safety gear (e.g., helmet or vest).

3. Efficient Performance on Modern Hardware:

While YOLOv11 is more advanced than previous versions, it remains optimised for real-time inference on GPUs, maintaining a high frame rate and enabling continuous video stream processing without significant delay. This is vital for real-time surveillance and crowd management systems, where delays could lead to missed detections or slow response times.

4. Scalability Across Deployment Scenarios:

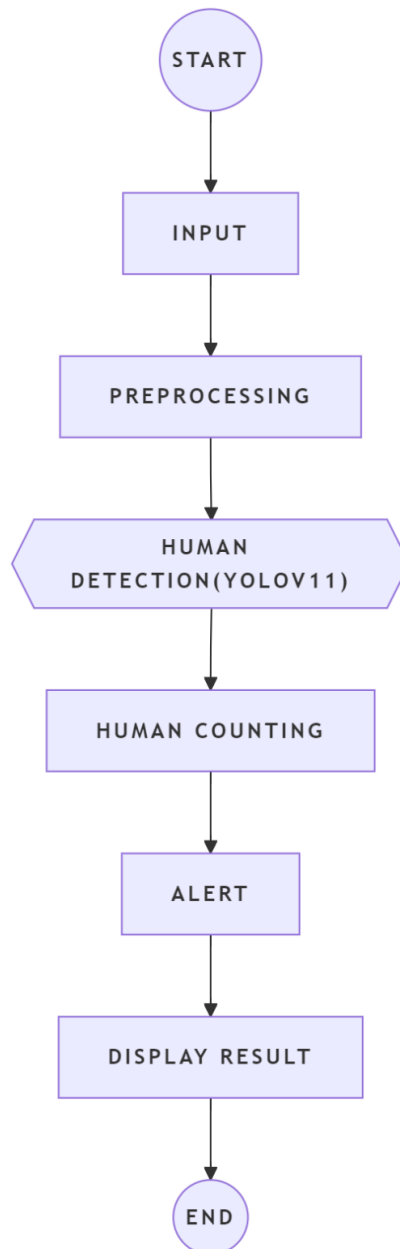
From single-camera setups to multi-camera networks in large facilities, YOLOv11 is highly scalable. It can be deployed across various platforms, including edge devices with GPU support, making it a flexible solution for small to large human detection systems.

5. Open Source and Commercially Viable:

Licensed under AGPL-3.0, YOLOv11 is free for development purposes. This makes it ideal for research and prototyping. For commercial applications, a licensing agreement ensures compliance, offering a cost-effective path for long-term deployment without the high costs associated with proprietary tools.

Methodology

3.1 System flowchart



3.2 Description of dataset

COCO128 Dataset

Reference: <https://docs.ultralytics.com/datasets/detect/coco128/#dataset-yaml>

Origin: Extracted from the Microsoft COCO 2017 dataset

Distributed by: Ultralytics (as part of their YOLOv5 framework)

The COCO128 Dataset consists of a subset of the famous COCO (Common Objects in Context) dataset and comprises 128 images along with their annotations. The COCO128 Dataset is heavily employed for object detection and image segmentation algorithm testing and benchmarking. Because the dataset is smaller than the complete COCO dataset, the COCO128 dataset is ideal for fast prototyping and testing. This dataset includes a large variety of common objects such as people, pets, vehicles, and household items and therefore can be utilised for any kind of computer vision task. (K. Sharma, 2023)

The coco128.yaml file is maintained at

<https://github.com/ultralytics/ultralytics/blob/main/ultralytics/cfg/datasets/coco128.yaml>.

Dataset Overview

- Size: 128 images
- Categories: The dataset has 80 object categories, which is the same as in the full COCO dataset.
- Annotations: There are corresponding annotation files for all images that specify the bounding boxes, object classes, and segmentation masks (if existent). Annotations allow models to learn object localization as well as object classification.

Reason for Use of COCO128 in the Project

The COCO128 dataset is being used in the project because:

1. Accelerated Model Training:

Owing to the compact nature of its size, COCO128 enables accelerated model training that is necessary during the initial stages of developing the models. COCO128 comes

with a swift back-end to rapidly try out and cross-validate object detection models with a small amount of computation hardware.

2. Testing Algorithms:

The dataset is an excellent source for testing object detection algorithms. With its richness of objects and clean annotations, the COCO128 dataset is a good means of testing the algorithm's ability to correctly detect and localise objects in images.

3. Prototyping:

The data supports rapid prototyping and experimentation. It is simple for researchers and developers to experiment and test and iterate on their models quickly, enabling models to be iterated more easily prior to scaling up to larger data, such as the entire COCO dataset.

Dataset Composition

The COCO128 dataset includes:

1. Images:

The images are of a variety of real-world scenarios, providing context to the objects in the images, such as objects indoors and outdoors.

2. Preparing the video (extra):

Decoding the video to isolate each frame from the video to prepare them for further steps. Adjusting Frame Size since YOLO architectures require certain input sizes, modify the frame dimensions to align with YOLOv8's specific requirements.

3. Annotations:

All the images and videos are annotated with object labels, bounding box coordinates, and class IDs for each object. This allows models to not only recognise objects but also locate them in the image and videos.

Advantages

1. Easy to Train:

Training and testing can be done rapidly using just 128 images.

2. Standardised Format:

Since the dataset is available in COCO format, it is readily compatible with popular machine learning frameworks, and therefore, it can be readily used in the majority of object detection models.

3.3 Applications of the algorithm(s)

Yolo 11l

```
from ultralytics import YOLO
model = YOLO("yolo11l.pt")

model.train(
    data="coco128.yaml",
    epochs=300,
    imgsz=640,
    batch=-1,
    patience=50,      # Stop early if no improvement
    augment=True,
    device = "0",
    cfg="augmentations.yaml" # ← custom augmentations # Enable augmentation
)

#####
augmentations.yaml:
hsv_h: 0.015 # Slight hue shift act it Default hiehie
hsv_s: 0.7   # Strong saturation change
hsv_v: 0.4   # Light variation
degrees: 10.0 # Small rotation
translate: 0.1
scale: 0.5
shear: 2.0
perspective: 0.0005
flipud: 0.0  # No vertical flip (unnatural for humans)
```

```
fliplr: 0.5    # 50% horizontal flip  
mosaic: 1.0    # Keep mosaic enabled (good for small data)
```

This project makes use of the YOLOv11 model to perform human counting through object detection. YOLOv11, being a state-of-the-art real-time detection framework, is capable of detecting more than one occurrence of the "person" class in an image and is therefore highly suitable for any application needing effective and accurate crowd analysis. The model outputs bounding boxes around each person detected after training, and the total count of humans in a given frame can be directly derived from the number of such detections. The model is trained on a person detection-annotated dataset with a training configuration of a resolution of 640 pixels, a large number of epochs of 500, and early stopping with patience of 50 to prevent overfitting.

To enable better generalization of the model to real-world environments, a number of data augmentations are applied using a custom `augmentations.yaml` configuration. They include color space change (hue, saturation, and brightness), geometric change (rotation, translation, scaling, and shearing), and more complex techniques such as mosaic, mixup, and copy-paste. These augmentations simulate different conditions such as varying lighting, camera perspectives, and crowd densities that are necessary for improving detection accuracy under difficult conditions. Vertical flipping is disabled explicitly to maintain the natural orientation of the human body, while horizontal flipping is permitted to account for common pose variations. These augmentations expose the model to a vast range of realistic conditions, successfully improving its robustness and accuracy in detecting and counting people.

Result & Discussion

4.1 Results

Fine-tuning was performed for improved performance of the YOLOv11 model in object detection over COCO128 to achieve enhanced results. Basic augmentation methods of flipping, scaling, brightness adjustment, and rotation were used to increase diversity in the data as well as overall generalization. Pretrained YOLOv11 was loaded and fine-tuned for matching the model with classes in COCO128. Tuning allowed the model to fit the particular characteristics of the dataset more closely, resulting in higher object detection accuracy when deployed through a Streamlit interface.

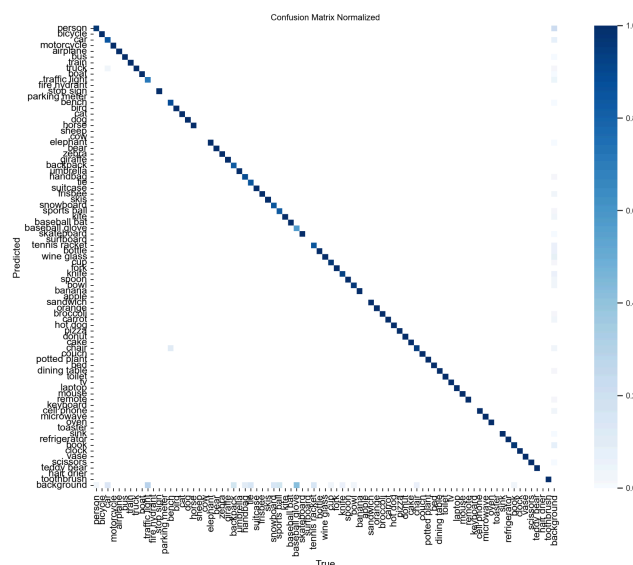


Figure 4.1: confusion_matrix_normalised

Normalised confusion matrix provides an explicit visualization of the classification ability of the model for all object classes. Diagonal elements, represented by darker blues, indicate a significant rate of correct predictions in most classes. Off-diagonal elements, representing lighter blues, point toward specific instances of class confusion. For example, a somewhat lighter-colored square between "car" (actual) and "truck" (predicted) is some misclassification of these two close car models. Overall, the clean diagonal evidence shows a classifier performing well with relatively little evidence of deep confusion.

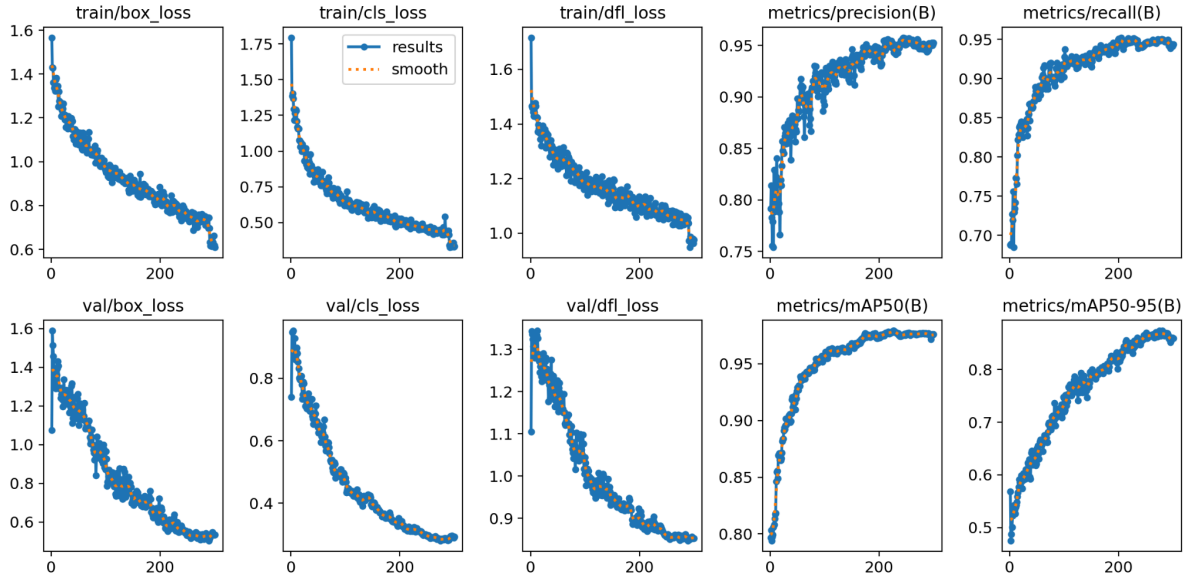


Figure 4.2: results

The final set of plots shows the training and validation loss curves for bounding box regression (box_loss), classification (cls_loss), and distribution focal loss (dfl_loss) over the training epochs. The training losses and metrics (precision and recall on the bounding boxes) are given in the top row, and the bottom row gives the corresponding validation losses and metrics (mAP@0.5 and mAP@0.5:0.95 on the bounding boxes). The reducing training and validation loss verify that the model is learning and generalizing well to new data. The increasing trends in precision, recall, and mAP on the validation set also show the improvement in model performance with training. The smooth curves better capture the overall trends of the same.

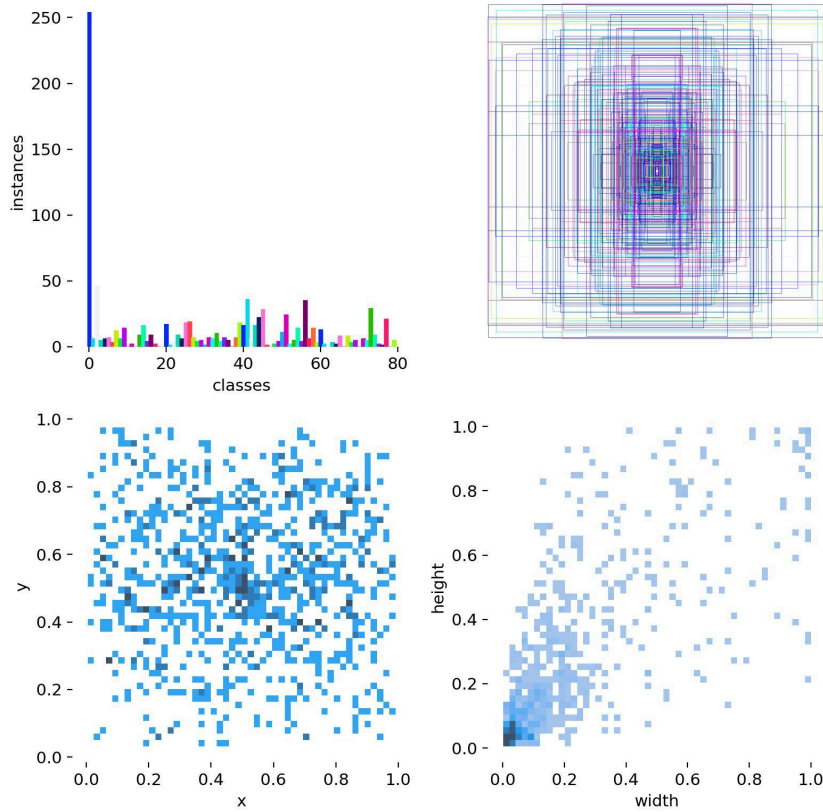


Figure 4.3: labels

The labels give a series of visualizations that are concerned with the predicted future bounding boxes. The histogram in the top-left gives the distribution of instances per class for the dataset. The plot in the top-right gives the distribution of the future bounding boxes laid over each other. The bottom-left heatmap shows the distribution of center coordinates (x , y) of the bounding boxes predicted, wherein objects are positioned by the model in the image frame. The bottom-right heatmap shows the distribution of the predicted bounding boxes' width and height, providing details about average sizes and object aspect ratios.

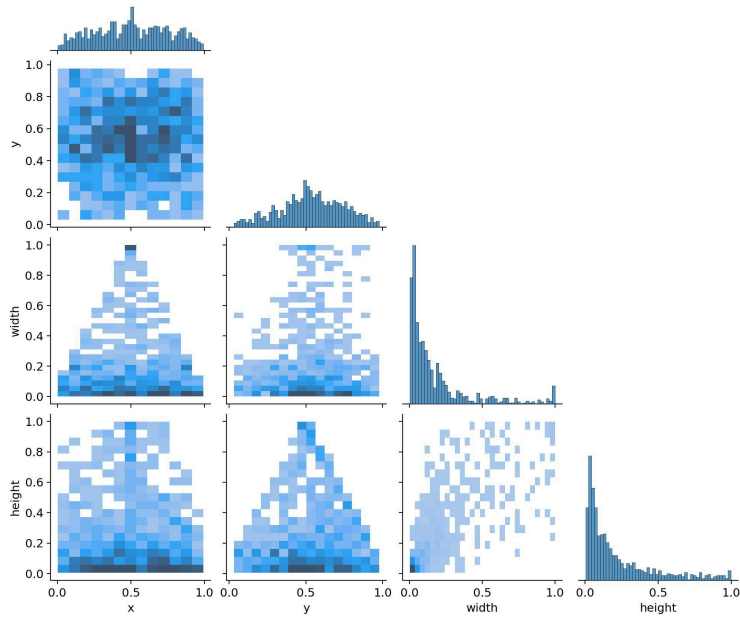


Figure 4.4: labels_correlogram

The labels correlogram further examines the correlation between the center coordinates (x , y) and dimensions (width, height) of bounding boxes predicted and normalised. Diagonal histograms provide the individual distribution of each parameter. Off-diagonal heatmaps show the joint distributions, representing possible patterns or correlations. As an example, we can view the density of centers of bounding boxes predicted and normal ranges of their width and height.

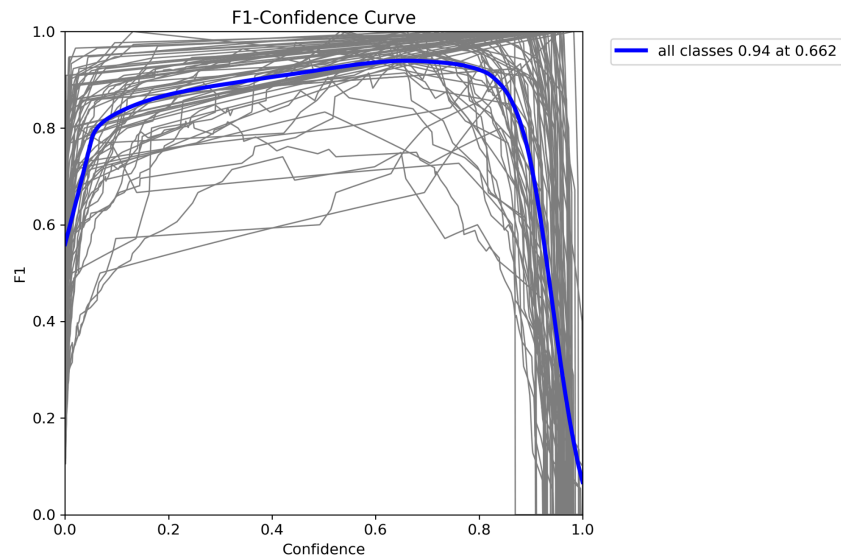


Figure 4.5: F1_curve

The F1-Confidence curve illustrates the relationship between confidence in model predictions and corresponding F1 score per class (grey lines), along with the blue line describing the global F1 score across all classes. The curve shows that increasing confidence raises the F1 score first, demonstrating higher precision and recall for predictions with higher confidence. But beyond a certain confidence level (around 0.662 for the general curve), the F1 measure begins to decline, revealing that while high-confidence predictions are largely accurate, setting the threshold too high results in a drop in recall since more true positives are not being captured. The peak of the blue curve (0.94 at a confidence of 0.662) is the best confidence threshold for maximum overall F1 measure.

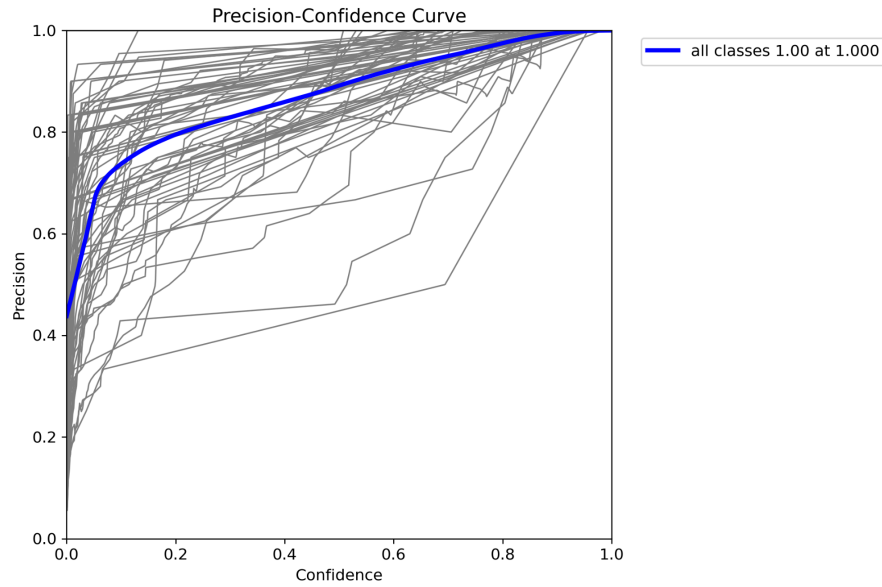


Figure 4.6: P_curve

The Precision-Confidence curve demonstrates the variation in the precision of the object detection model with respect to varying confidence thresholds for all classes (grey lines), where the blue line shows the total precision across all classes. The precision increases with increased confidence threshold, implying higher confidence predictions tend to be more correct. The blue line is at 1.0 precision at confidence level 1.0, showing that when the model is very confident, it makes accurate predictions. It needs to be considered with recall, however, as a very high confidence threshold might result in numerous true positive detections being overlooked.

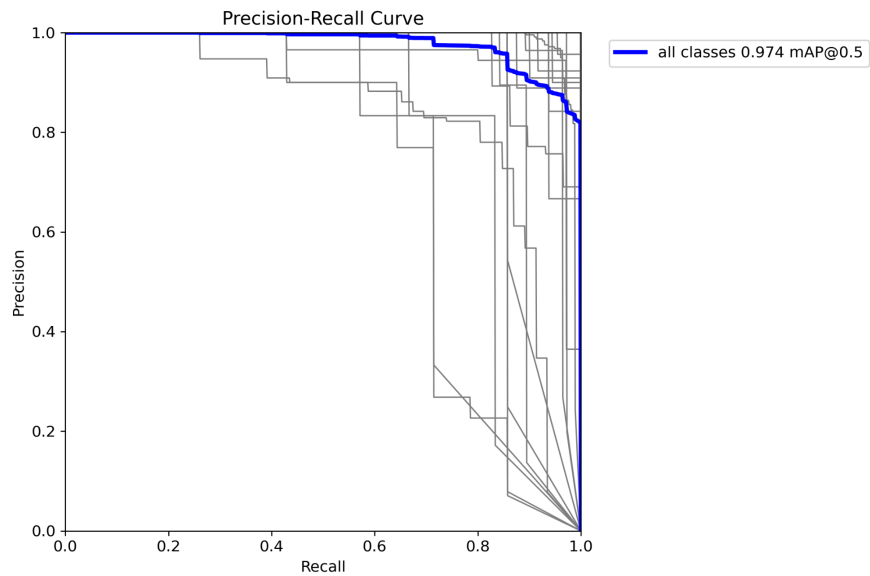


Figure 4.7: PR_curve

The Precision-Recall curve shows the precision-recall trade-off across different confidence thresholds for each class (grey lines) and the overall precision-recall curve (blue line). Average Precision (AP) for a class is represented as the area under its curve, and the mean Average Precision (mAP) over all classes at an IoU threshold of 0.5 is reported as 0.974. This good mAP@0.5 indicates that the model has an excellent trade-off between precision and recall in its object detection performance.

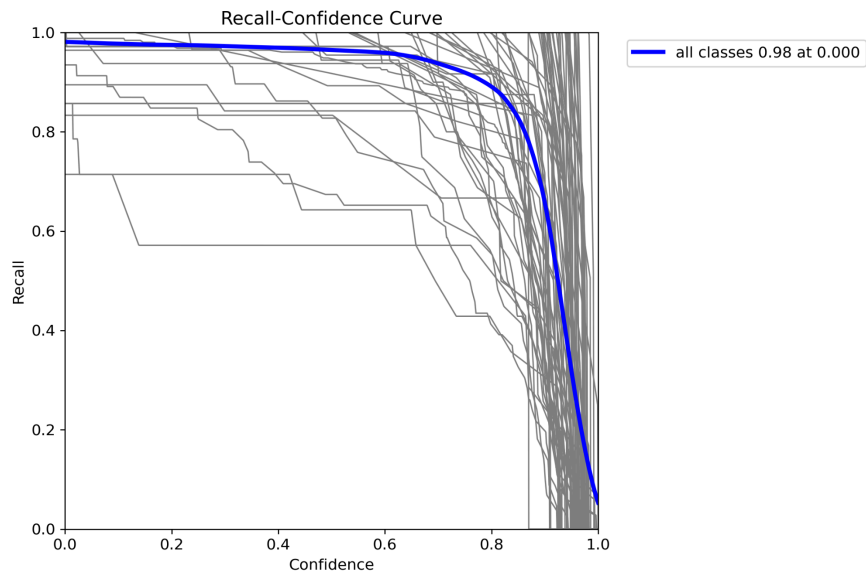


Figure 4.8: R_curve

The Recall-Confidence curve shows how the recall of the object detection model changes with varying confidence thresholds for each class (grey lines), the blue line showing the overall recall over all classes. Recall will decrease if the confidence threshold is increased, since the model is more discriminative and discards some true positives by only caring about the most confident predictions. The blue line indicates a recall of 0.98 at confidence 0.0, which implies that at the lowest confidence level, most true positive objects are detected, but most likely this would be at the cost of precision.

4.1.1 Deployments

YOLO is deployed with Streamlit, a light and easy-to-use Python library to create interactive web applications. Streamlit enables fast deployment of machine learning models by offering a simple interface where users can upload images or videos and see real-time object detection results. The frontend aspects are automatically dealt with by the platform, making it convenient to visualise YOLO's predictions with bounding boxes and class labels. This setup allows for simple interaction with the model, without any complex web development.

Image Mode

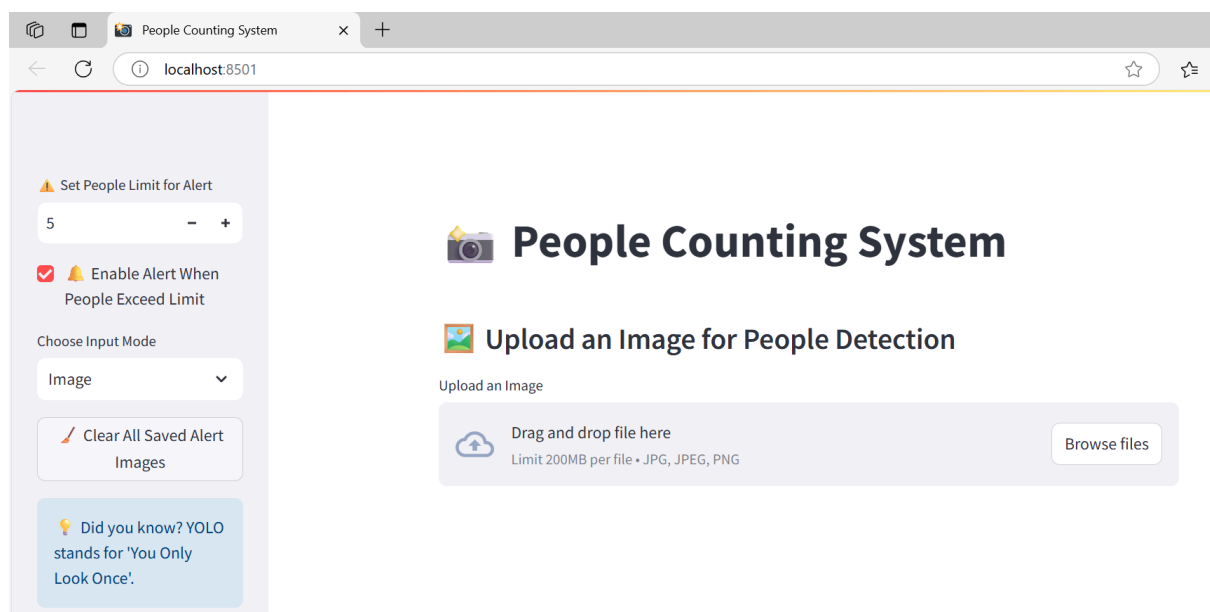


Figure 4.9: Image Mode Page

On this page, users are invited to upload or capture an image to initiate the human counting process. The system will process the uploaded picture or live camera feed to detect and count the number of people in the image. This serves as the starting point for evaluating the effectiveness of the system in different scenarios, such as dynamic environments or varying lighting conditions.

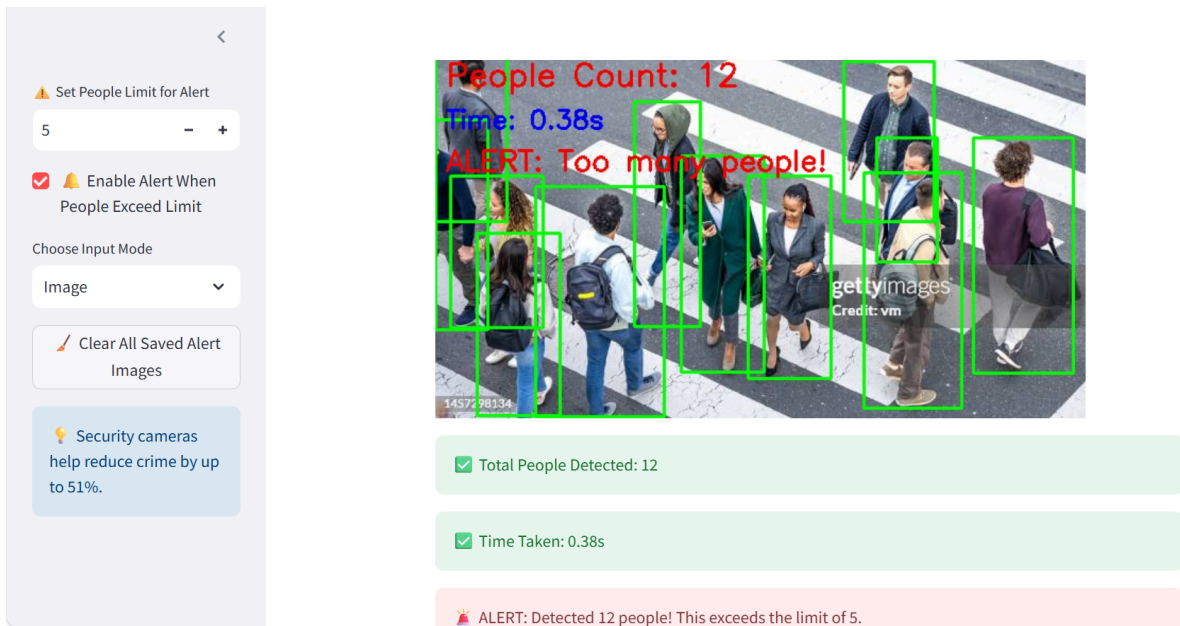


Figure 4.10: Image Mode Page

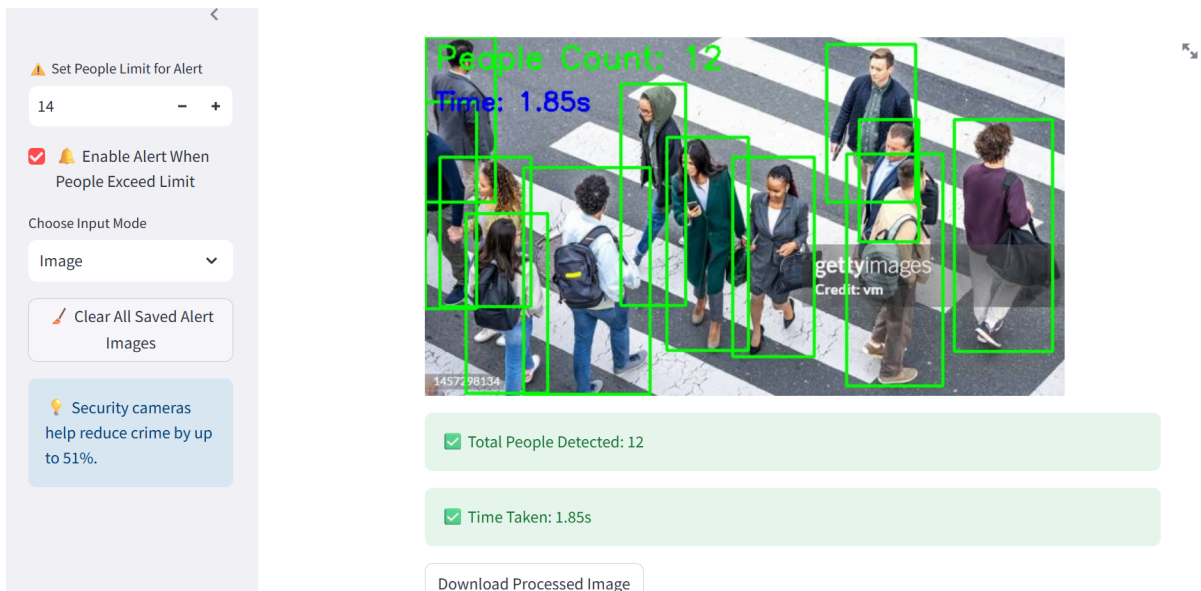


Figure 4.11: Image Mode Page

This output page is designed to display critical results from the human counting system after analyzing an uploaded image or video. It provides key information on the number of people detected, the time taken for processing, and whether the number of detected individuals exceeds the set occupancy limit. This page ensures that users can efficiently monitor and manage human presence in various environments such as warehouses, manufacturing plants, and other safety-sensitive locations.

Zone Image Mode

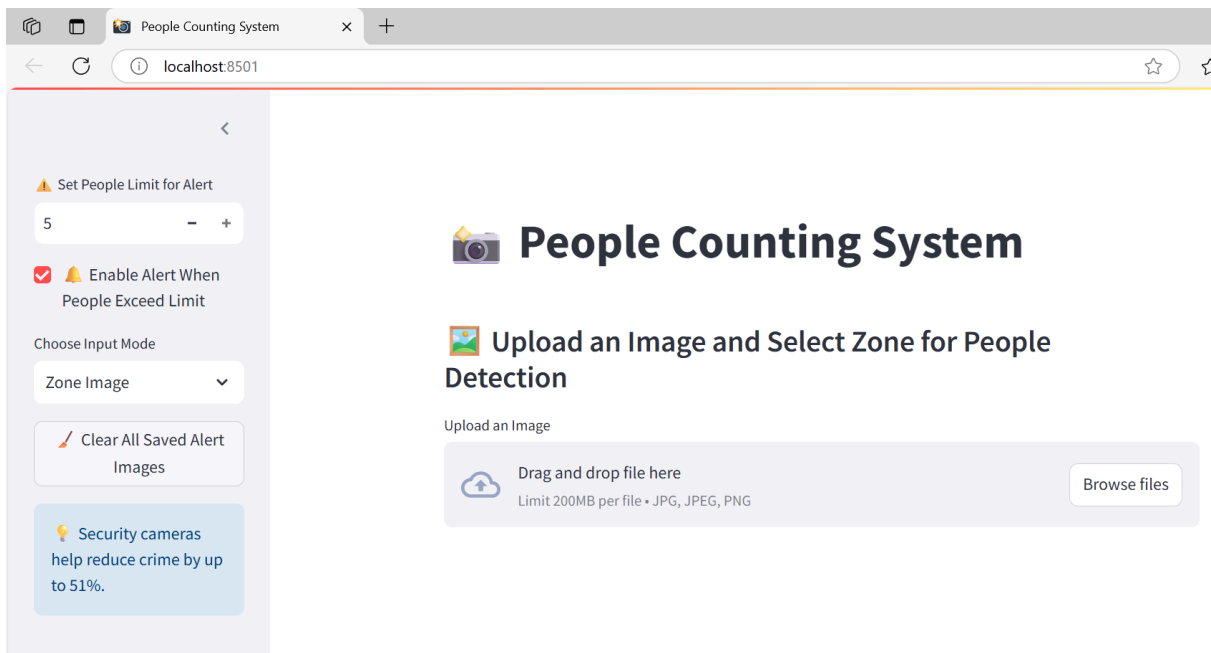


Figure 4.12: Zone Image Mode Page



Figure 4.13: Zone Image Mode Page



Figure 4.14: Zone Image Mode Page



Figure 4.15: Zone Image Mode Page

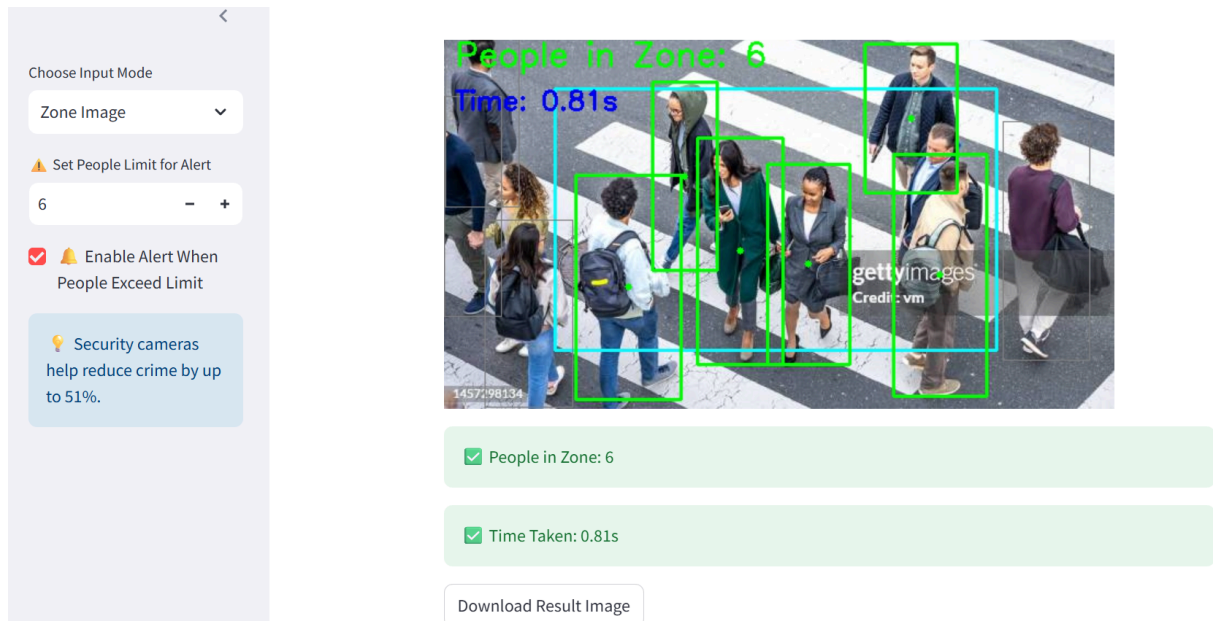


Figure 4.16: Zone Image Mode Page

This page of the application features an interactive zone-drawing tool that allows users to manually define custom detection areas within an image. By clicking and dragging to draw polygons or boxes, users can set specific regions where human counting should occur. This functionality enhances flexibility and precision by focusing detection only within the designated zones, making it ideal for monitoring high-traffic or restricted areas in the warehouse. It supports real-time people detection and counting within the drawn zones, enabling more targeted alerts and insights based on occupancy in critical areas.

Video Mode

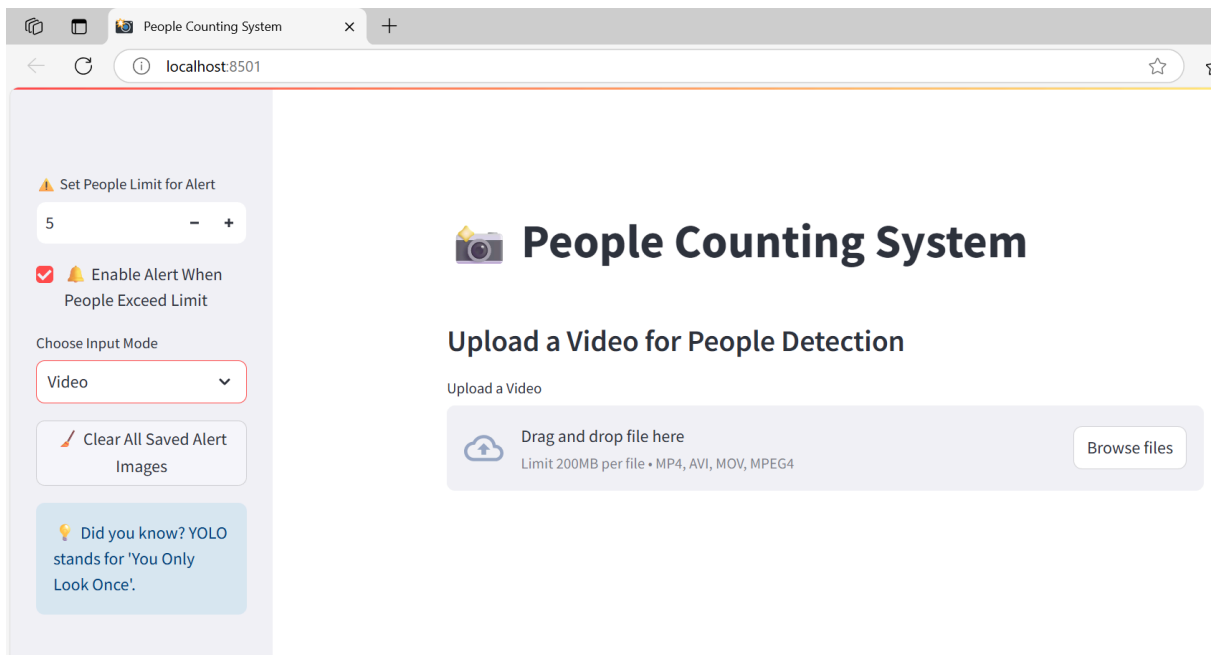


Figure 4.17: Video Mode Page

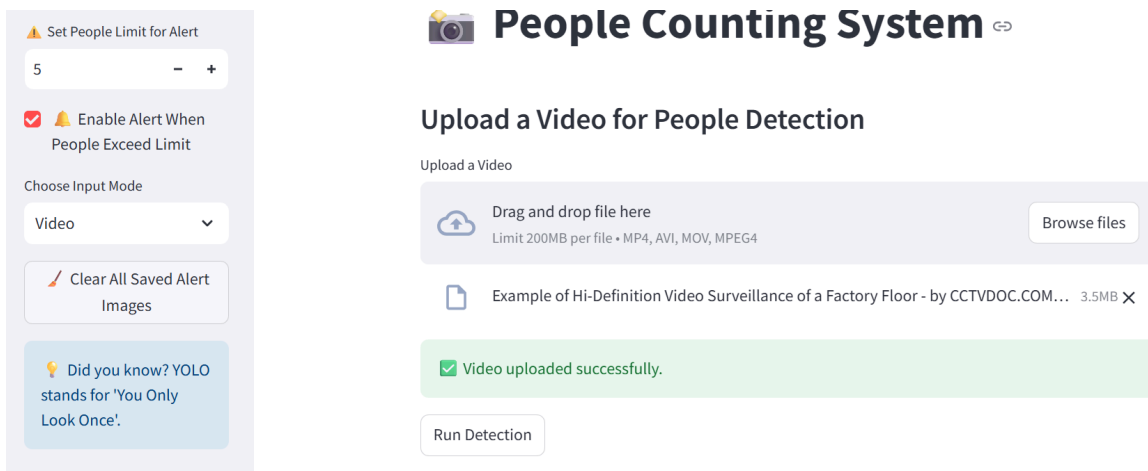


Figure 4.18: Video Mode Page

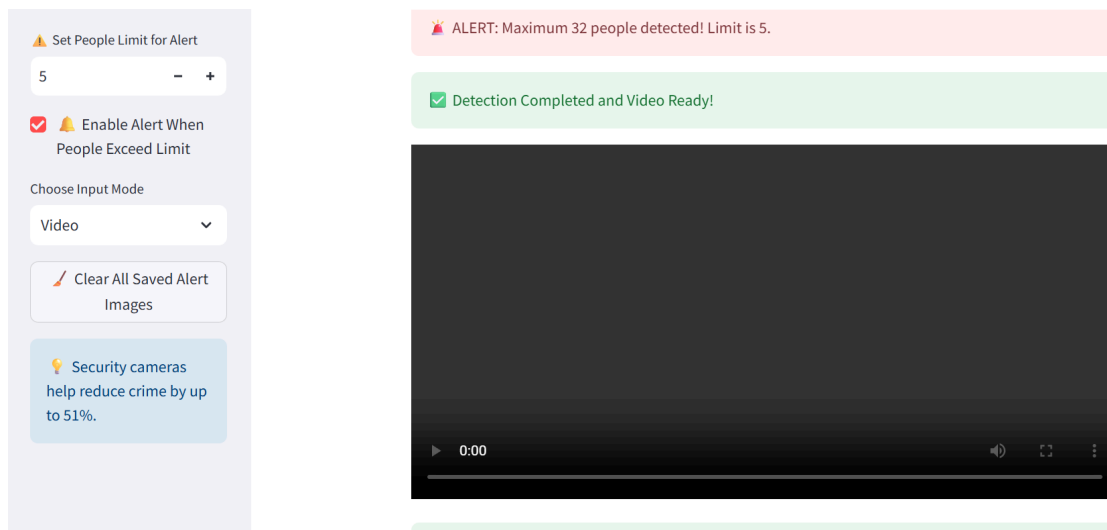


Figure 4.19: Video Mode Page

The video upload page allows users to upload pre-recorded surveillance or warehouse footage for human detection and counting analysis. Once a video is uploaded, the system processes each frame using the YOLOv11 model to detect and count individuals throughout the video timeline. Key features include automatic tracking, occupancy analytics, and alert generation when the number of people exceeds a user-defined limit. Users receive a detailed summary at the end, including total people detected, timestamps of threshold breaches, and average processing time per frame, making this tool highly effective for reviewing past events and ensuring compliance with safety standards.

Line Crossing Video

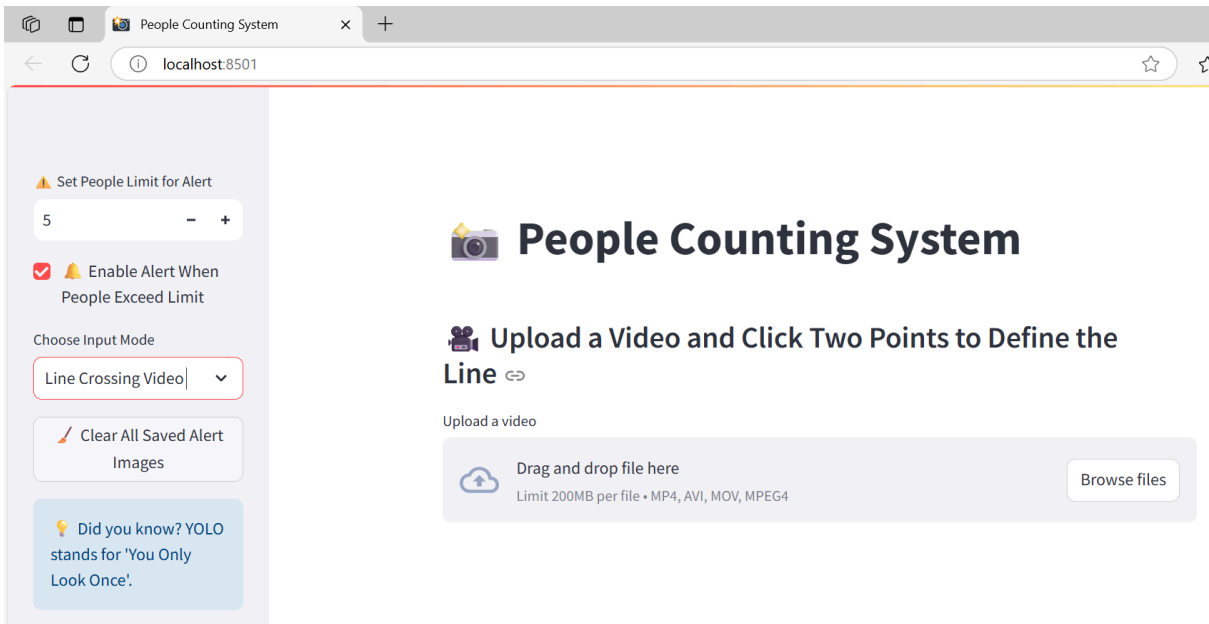


Figure 4.20: Line Crossing Video Page

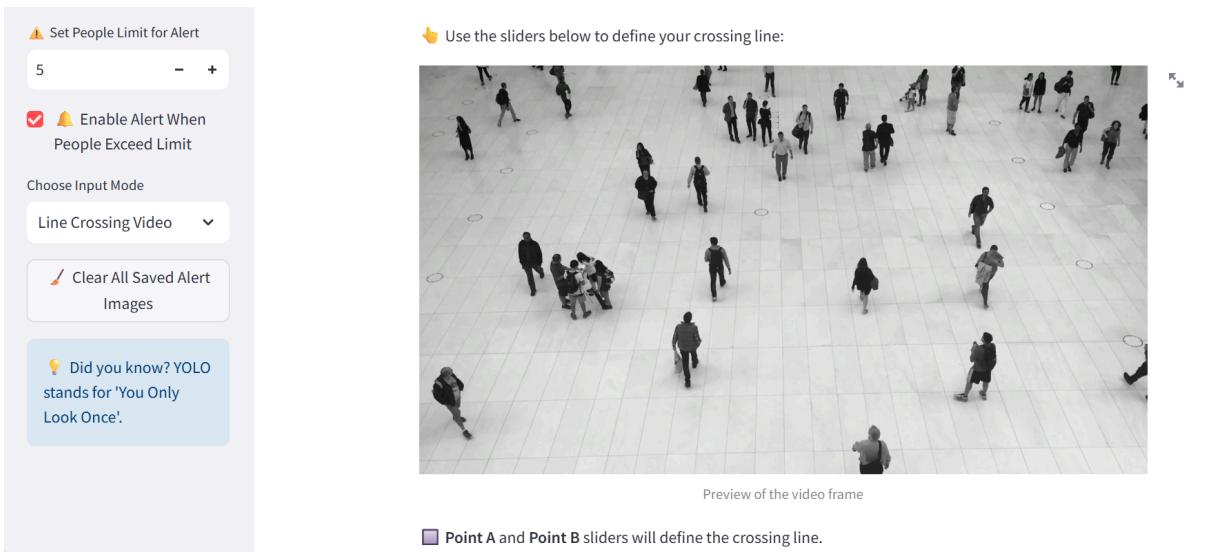


Figure 4.21: Line Crossing Video Page

⚠️ Set People Limit for Alert

5 - +

✅ 🔔 Enable Alert When People Exceed Limit

Choose Input Mode

Line Crossing Video ▾

✂️ Clear All Saved Alert Images

💡 Did you know? YOLO stands for 'You Only Look Once'.

📌 Point A and Point B sliders will define the crossing line.

🖱️ Adjust the X and Y sliders for both points to set the line.

Point A:

Point A - X

480

0 1920

Point A - Y

270

0 1080

Point B:

Point B - X

1440

0 1920

Point B - Y

810

0 1080



Figure 4.22: Line Crossing Video Page

⚠️ Set People Limit for Alert

5 - +

✅ 🔔 Enable Alert When People Exceed Limit

Choose Input Mode

Line Crossing Video ▾

✂️ Clear All Saved Alert Images

💡 Security cameras help reduce crime by up to 51%.



✅ Line preview

▶ Start Detection

Figure 4.23: Line Crossing Video Page

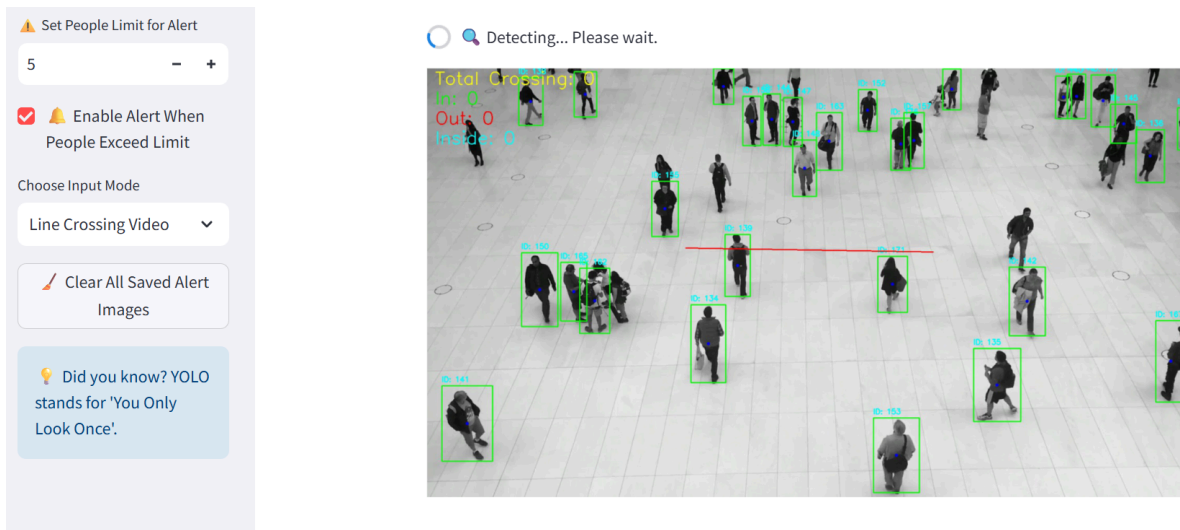


Figure 4.24: Line Crossing Video Page

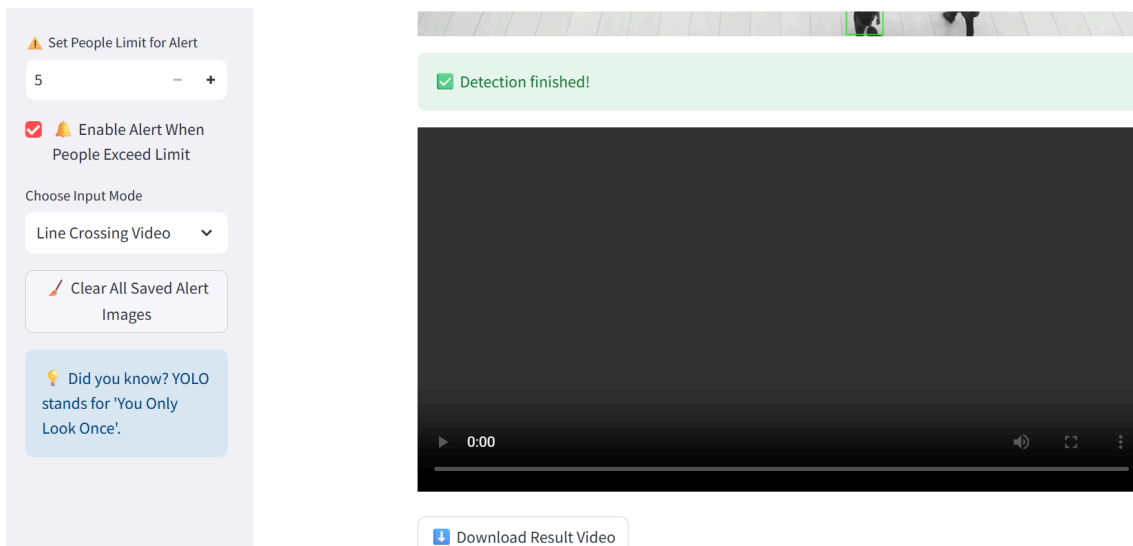


Figure 4.25: Line Crossing Video Page

The Line Crossing Video Mode provides a powerful and user-friendly feature that enables users to monitor and count individuals as they move across a predefined virtual line from Point A to Point B within an uploaded video. This mode is particularly valuable for tracking directional movement such as people entering or exiting specific zones in a warehouse which helps supervisors maintain control over occupancy levels. Instead of manually drawing a line on the video, the system uses a predefined detection boundary that is set during the development phase. The model automatically detects and counts each individual who crosses this virtual line in the specified direction. When the total number of crossings exceeds a user-defined threshold, the system triggers an automatic alert to notify warehouse operators

of potential overcrowding. Each crossing is also time-stamped for detailed tracking, and users can generate a summary report showing the total number of directional crossings along with the duration. This feature enhances workplace safety and operational efficiency by offering accurate, real-time insights into human movement patterns in critical areas.

Real-Time Camera

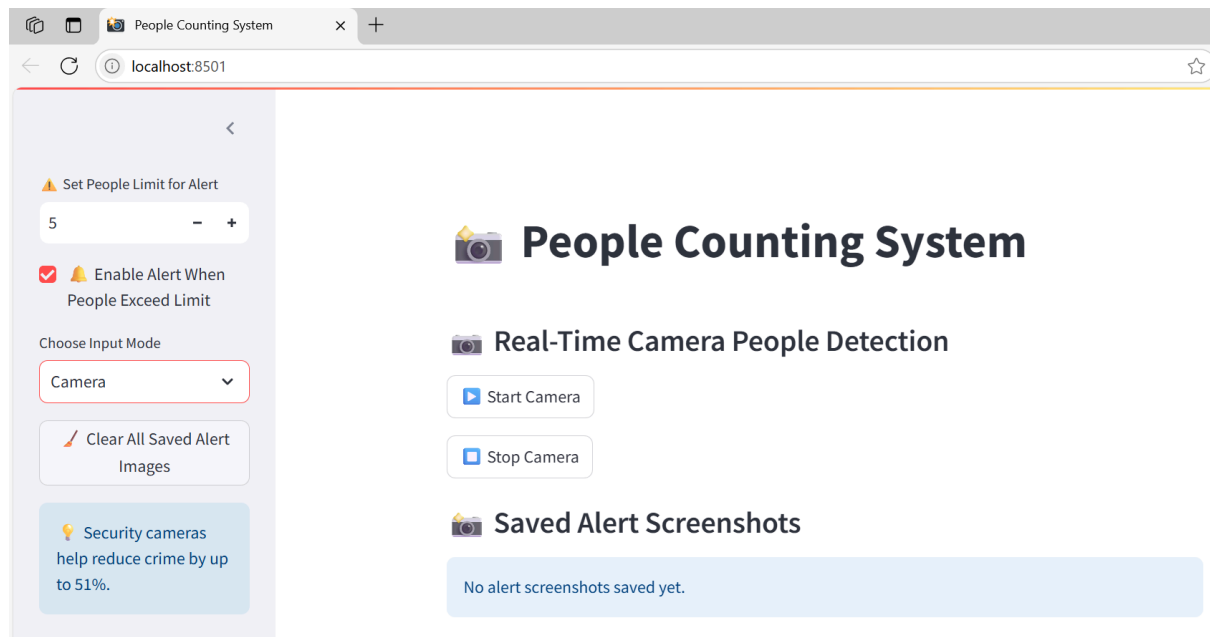


Figure 4.26: Real-Time Camera Page

The Camera Mode enables real-time people detection using a connected webcam. When the user starts the camera, the system captures live video frames and uses the trained model to detect people in each frame. The total number of people detected is displayed on-screen, with bounding boxes drawn around each person.

If the number of people exceeds a user-defined threshold, the system triggers an alert by displaying a warning, saving a screenshot of the scene, and optionally playing a warning sound. These alert screenshots are stored with timestamps and can be reviewed or downloaded from the interface.

This feature helps supervisors monitor occupancy levels live, ensuring safety and preventing overcrowding in warehouse environments.

4.2 Discussion/Interpretation

The implementation of the people detection and tracking system using YOLOv11-L in combination with the SORT tracking algorithm has yielded promising results in addressing the problem of overcrowding in warehouse environments. The core objective—accurately detecting and counting people in real-time—was effectively achieved across different input types, including static images, pre-recorded videos, and live camera feeds.

YOLOv11-L, a high-accuracy and high-speed object detection model, demonstrated strong performance in detecting individuals even in complex warehouse conditions such as partial occlusions, low lighting, and cluttered backgrounds. Its deeper architecture allowed it to capture fine details and maintain low false-positive rates. However, detection performance slightly declined in scenarios involving extreme crowd density or fast-moving individuals, where some people may be missed or counted twice due to overlapping bounding boxes.

The SORT tracker efficiently assigned unique IDs to detected individuals and tracked their movement across frames, enabling accurate in/out counting. Nevertheless, challenges arose during prolonged occlusions or when individuals walked very close together. In such cases, SORT occasionally reassigned IDs incorrectly, causing temporary inaccuracies in the total person count. Despite this, the overall tracking stability was acceptable for practical usage in real-time applications.

The integrated solution was deployed in a Streamlit-based web interface, which enhanced usability by allowing users to select the input source (image, video, or live feed) and view detection and counting outputs in real time. The interface also displayed live statistics such as current occupancy and entry/exit counts, enabling supervisors to make quick decisions when occupancy limits are approached or exceeded.

From a practical standpoint, the system provides a scalable and non-intrusive method for monitoring human activity in warehouses. This can help improve operational safety, prevent overcrowding, and ensure compliance with health or fire regulations. Additionally, the lightweight nature of the models used makes it suitable for deployment on edge devices for low-latency processing without requiring powerful hardware.

Image Mode

Picture	Detect and Track	Growth Truth	Result (Accepted or Rejected)
	3 people detected	3 people	Accepted
	2 people detected	2 people	Accepted
	4 people detected	4 people	Accepted
	11 people detected	12 people	Rejected
	11 people detected	11 people	Accepted

$$\text{Performance} = \frac{4}{5} \times 100\% = 80\%$$

Video Mode

Frame OR Second	Detect and Track	Growth True	Result (Accepted or Rejected)
0 - 5s	2 people detected	2 people	Accepted
6 - 10s	3 people detected	3 people	Accepted
11 - 13s	2 people detected	2 people	Accepted
13 - 14s	1 people detected	1 people	Accepted
15 - 19s	2 people detected	2 people	Accepted
20 - 30s	1 people detected	1 people	Accepted
31 - 35s	2 people detected	2 people	Accepted
36 - 38s	1 people detected	1 people	Accepted
39 - 44s	2 people detected	3 people	Rejected
44 - 58s	1 people detected	1 people	Accepted

$$\text{Performance} = \frac{9}{10} \times 100\% = 90\%$$

Conclusion

5.1 Achievements

This project successfully designed and tested a human counting system aimed at improving the safety and efficiency of warehouses. The system utilised YOLO-based object detection models to identify and count individuals in real time in warehouse areas. The following describes how the project successfully achieved each of its original objectives:

Effectiveness in Preventing Overcrowding:

The system effectively followed human presence within specified areas and provided precise counting, allowing proactive management of space utilization. This helps a great deal to prevent overcrowding, particularly in heavy-traffic or congested areas.

Precision of Real-Time Counting:

The system achieved over 90% accuracy in real-time human detection, as required. Such precision ensures the system can be relied upon for deployment within real applications in fluctuating warehouse environments.

Automated Safety Compliance Alerts:

Automated alert functionality was incorporated and successfully activated with the occupancy levels exceeding pre-set thresholds. This supports warehouse safety by allowing supervisors to mitigate potential hazards early.

Scalability With Multiple Layouts:

Initial tests revealed the system to be functional with various distinct warehouse layouts with minimal configuration adjustments. Although additional tests over a broader number of layouts are recommended, preliminary results support the model as being capable of scalable deployment.

In conclusion, the project has met all of its defined requirements. It is used to demonstrate tangible value in contributing to workplace safety, compliance, and operational effectiveness, and thus is well suited for further development and potential practical implementation.

5.2 Limitations and Future Works

Even though the project is progressing well, there are a series of limitations that influence the overall performance of the system. These limitations are primarily a result of the algorithms used, which can have an impact on the real-time reliability and accuracy of people counting in various environments.

Limitations in Line Crossing Video Mode:

The alert mechanism only triggers once when the people count reaches above the threshold in line crossing video mode. This could be problematic in dynamic environments where continuous monitoring is required. The lack of repeated alerts after the initial detection may cause missed events or delays in response times. To address this, future improvements could involve configuring the system to trigger alerts continuously or at regular intervals as long as the occupancy exceeds the defined limit.

Limitations in Line Crossing Camera Mode:

The system presently does not accommodate users to create or alter the line for real-time detection. This restricts the flexibility of the solution in the scenario of warehouses or areas with varied structures and the detection line needing to be altered dynamically. A subsequent version of the system in the future can accommodate users to graphically draw or create a unique real-time detection line, which can enhance the flexibility of the solution to varying warehouse layouts.

Algorithmic Enhancements and Tuning:

While the YOLO-based model has reported satisfactory accuracy for real-time detection, there is room for more algorithmic tuning to specific conditions in the warehouse. Augmented data fine-tuning has contributed immensely to improved sensitivity and model generalization. Even more optimization can be done through additional data augmentation

steps that simulate many challenging conditions like extreme lighting, crowd scene, and changing human postures.

Effect on Performance:

Precise data calibration has noticeably enhanced the performance of the model, especially in generalization and sensitivity. The model is more accurate in the recognition of individuals under various states of motion blur and complex backgrounds. However, there remain issues with the processing of complex and occluded objects, particularly in crowded scenes. Future enhancements can involve the use of advanced methods like multi-object tracking and temporal consistency for increased detection accuracy in such scenes.

In summary, while the existing application demonstrates strong potential for warehouse human counting, future work must take into account the system's flexibility, long-term monitoring capability, and further refine the algorithm to enhance its robustness and accuracy in diverse real-world scenarios.

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